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# METAPOPULATION DYNAMICS AND CONSERVATION: A SPATIALLY EXPLICIT MODEL APPLIED TO BUTTERFLIES

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**Abstract**

*We analyse in detail the metapopulation structure of three species of butterflies in regions where they are endangered—*Melitaea cinxia *in Finland and* Hesperia comma *and* Plebejus argus *in the UK. Metapopulations are assemblages of local populations in which the whole (metapopulation) may persist even if all the components (local populations) are vulnerable to extinction. Patterns of habitat patch occupancy and local density in the three species support the general predictions of metapopulation models. We develop a spatially explicit metapopulation model and fit it to data on the three species. The model is tested with independent data on* H. comma *and* P. argus, *for which it predicts the rate and pattern of spread into networks of vacant habitat patches following introduction or natural recolonization. The model can be used to assess the potential of specific networks of habitat patches to support viable metapopulations of given species, and it therefore has great potential value for conservation of butterflies and other species which occur as systems of interconnected small populations.*

*Keywords:* metapopulation, extinction, colonization, spatially explicit model, Britain, Finland.

## INTRODUCTION

Increasing fragmentation of natural habitats and their plant and animal populations has swelled the interest in metapopulation dynamics (Gilpin & Hanski, 1991). Levins (1970) defined a metapopulation as a 'population of populations', an assemblage of local populations connected to each other by occasional migration. As such, the whole (metapopulation) may persist even if all the components (local populations) have some probability of extinction. In a broader sense, any assemblage of local populations which are connected by migration may be called a metapopulation, regardless of the frequency of local extinctions. However, if the migration rate is so high that individuals from different habitat patches become entirely mixed with each other there is no need for the metapopulation concept.

Compared with the conventional emphasis on the dynamics of local populations, metapopulation dynamics direct attention to the significance of the processes of local extinction, migration among local populations, and (re)colonization of vacant habitat patches in the long-term persistence of a species. Metapopulation dynamics also draw attention to the role of currently unoccupied but suitable habitat patches.

The metapopulation concept has been considered appropriate for various plants (Hanski, 1982; Collins & Glenn, 1990, 1991), amphibians (Sjögren, 1991), birds (Van Dorp & Opdam, 1987; Verboom *et al.*, 1991) and mammals (Smith, 1974, 1980; Peltonen & Hanski, 1991; Verboom *et al.*, 1991; Hanski, 1992) inhabiting fragmented landscapes, and butterflies are no exception (Harrison *et al.*, 1988; Harrison, 1989; C.D. Thomas & Harrison, 1992). In the past, the emphasis in butterfly conservation has been to reduce the rates of local extinction by maintaining local habitat quality (J. A. Thomas, 1984, 1991). But butterfly populations rarely persist for long, say more than 100 years, in single, isolated habitat patches (C.D. Thomas *et al.*, 1992). The emphasis in conservation must therefore eventually expand from maintaining the specific requirements of a particular species in single habitat patches to maintaining networks of such patches, which can be connected by occasional migration (C. D. Thomas *et al.*, 1992; C. D. Thomas, 1993; Warren, 1993*b*).

The purpose of this paper is to ask when is it necessary to expand the perspective from the level of local populations to the level of metapopulations, and to describe a simulation model that can be applied to specific metapopulations. The model allows us to

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estimate the long-term probability of metapopulation persistence in natural networks of habitat patches. We use a simulation model because most of the theoretical results and predictions of analytical models are qualitative, and hence difficult to apply to specific localities and specific metapopulations. We demonstrate the use of the simulation model with three examples from our research.

## BUTTERFLY METAPOPULATIONS: CONCEPTS AND DOMAIN OF APPLICATION

Two key concepts in the study of metapopulation dynamics are local population and habitat patch (for further discussion on concepts and terminology see Gilpin & Hanski, 1991). By *local population* we mean a group of individuals which live in the same place (within one habitat patch) and mate and interact with each other more or less randomly. This is similar to the concept of a 'closed' population, or colony (e.g. Ehrlich, 1961, 1965; J. A. Thomas, 1984; Warren, 1993*a*). By *habitat patch* we mean a continuous area, defined by the distribution of host plant(s) and other specific resource requirements, within which a local population may survive for many years, in principle indefinitely.

### Butterfly species with metapopulations

Forty-seven of the 60 British butterfly species are regarded as having 'closed colonies' (*sensu* Ehrlich, 1961, 1965), or well-defined local populations, in which numbers are determined more by local birth and death processes than by migration (J. A. Thomas, 1984; Warren, 1993*a*). Nonetheless, local populations are often situated sufficiently close to each other so that some migration between them does occur (Ehrlich *et al.*, 1975; Hanski *et al.*, in press), and natural local extinctions and colonizations have been observed (J. A. Thomas, 1984, 1991; C. D. Thomas & Harrison, 1992; C. D. Thomas *et al.*, 1992; C. D. Thomas, 1993; Warren, 1993*b*).

Some species with 'closed' local populations have habitat patches so densely packed that their distribution is practically continuous and there is virtually never any vacant habitat. In their case a metapopulation approach is not particularly helpful, though it could become so if the environment changes. For example, *Maniola jurtina* is very common in the UK, but has become endangered in, for example, Finland, apparently because the density of suitable habitat patches has drastically declined during past decades.

The remaining 13 species have 'open' or migratory population structures. Local abundances of these species are dominated by immigration and emigration, and local populations cannot legitimately be considered as becoming extinct, or empty habitat as being colonized.

Excluding species with open populations and those with closely-packed habitat patches, at least half of the British species appear to conform to a genuine metapopulation structure (below), and this includes almost all of the rarest species. A similar situation is likely to occur in other countries where natural habitats have become fragmented.

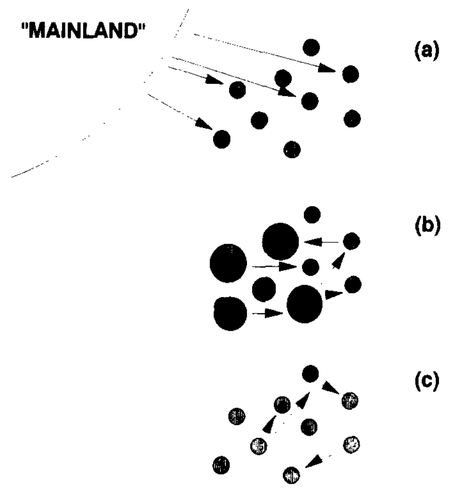



**Fig. 1.** A continuum of metapopulation structures, from a pure mainland–island structure (a), in which all migration occurs from the 'mainland' to the islands (habitat patches), to the 'Levins structure' (c), in which all habitat patches are of the same size and there is no outside migration. Most real metapopulations represent case (b), an intermediate metapopulation structure (from Hanski & Gyllenberg, 1993).

### Metapopulation structures

It is useful to make a distinction between two extremes of a continuum of possible metapopulation structures. One extreme is represented by the 'Levins model' (Levins, 1969, 1970), in which all habitat patches are of the same size and are so small that all local populations have a significant risk of extinction (Fig. 1(c)). The other extreme is the mainland–island metapopulation structure (Fig. 1 (a)), which was first explored by MacArthur and Wilson (1967) in the context of island biogeography. Here, one 'patch' (mainland) is so large that its local population (practically) never goes extinct, and is the only source of migrants to the remaining small habitat patches; this structure is also called the Boorman–Levitt metapopulation (Boorman & Levitt, 1973). Some butterfly metapopulations appear to have a 'mainland–island' structure (Harrison *et al.*, 1988; Harrison, 1989).

The present paper describes several examples of butterfly metapopulations which represent the intermediate case, Fig. 1(b), with spatial variation in patch sizes but no very large patches (mainland). Most real metapopulations of butterflies and other species are likely to be of this type. Clearly, metapopulation dynamics are especially critical to overall persistence in the cases where even the largest habitat patches are so small that their local populations have a significant risk of extinction.

### Risk of local extinction

Figure 2 shows how the persistence time of a model population depends on the equilibrium population size

**Table 1. Temporal variability in butterfly populations as measured by the standard deviation of $\log_e$-transformed yearly abundances, $s$**

The table also gives the median, minimum and maximum population sizes (*E. editha*) or population counts (*P. argus, H. comma*). $T$ is the length of the study period (years = generations).

| Species | $s$ | median | min | max | $T$ | References[a] |
|---|---|---|---|---|---|---|
| *Euphydryas editha* | 1·24 | 341 | 18 | 2 000 | 26 | 1 |
| | 1·26 | 267 | 40 | 7 227 | 26 | 1 |
| *Plebejus argus* | 0·70 | 347 | 76 | 825 | 12 | 2, 3 |
| *Hesperia comma* | 0·84 | 45 | 9 | 164 | 11 | 3 |

[a] 1, Harrison *et al.* (1991); 2, C. D. Thomas (1991); 3, C. D. Thomas (unpublished data).

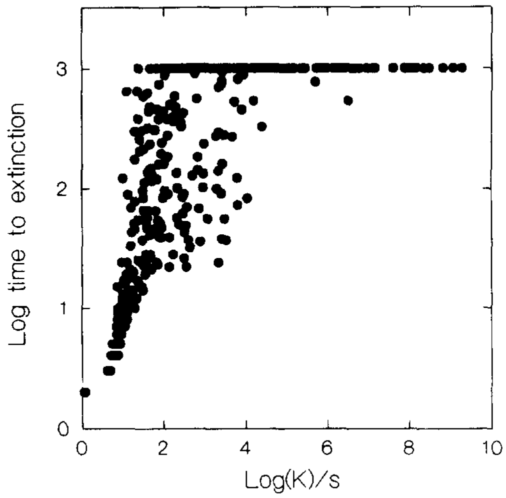
and on temporal variability in population size. The results were obtained from the model

$$N_{t+1} = N_t \exp\{r[1 - (\alpha N_t)^{\theta} + \varepsilon_t]\} + D_t \qquad (1)$$

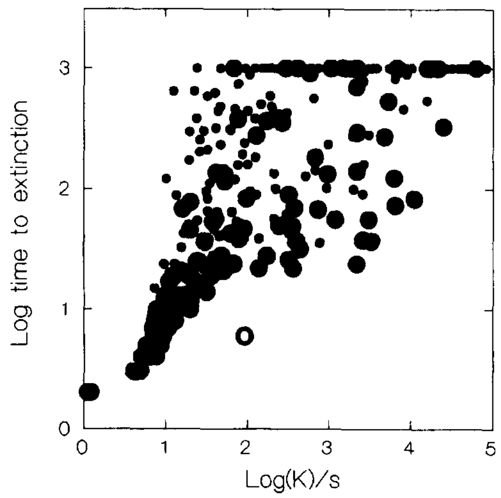
where $N_t$ is population size in generation $t$, $r$ is the intrinsic rate of population increase, $1/\alpha$ gives the equilibrium population size (which is stable for $r\theta < 2$), $\theta$ modifies the strength of density dependence near equilibrium (density dependence increases with increasing $\theta$), and $\varepsilon_t$ is a random normal variable with zero mean and variance of $\sigma^2$. $D_t$ is another random normal variable with zero mean and variance of $N_t$; $D_t$ was used to include in the model, albeit in a very nonmechanistic manner, the effect of demographic stochasticity.

In this model, the logarithm of the time to extinction of a local population is related to $\log(1/\alpha)/s$, where $s$ is the standard deviation of yearly log-abundances, estimated using the first 20 generations of the model output (Fig. 2).

Assuming that $s = 1$, which is a reasonable value for at least some butterflies (Table 1), the results in Fig. 2 suggest that any isolated population with the equilibrium population size <50 individuals will not survive for more than about 100 generations at most (x axis values <1·7 in Fig. 2), and typically such small populations would go extinct much sooner. On the other hand, many populations with the equilibrium size >500 could be expected to survive for more than 100 generations (x axis values >2·7), and populations of 100 000 for many hundreds of generations (x axis values >5). Though it is evident from Fig. 2 that no precise predictions about the probability of extinction of local populations can be made, we may conclude that population size should be many thousands of individuals before one could rest assured about its long-term survival, *assuming that the pattern of variability remains the same*. This final qualification is very important: if there are systematic changes in the environment, as has clearly been the case for most butterflies, long-term predictions based on past population performance cannot, and should not, be trusted. Figure 2 includes an empirical point which is based on data for *Plebejus argus* (C. D. Thomas, 1993). In this case the measured extinction rate is higher (expected time to extinction shorter) than predicted by the model. This observation underlines the point that extinctions may often be caused by 'extra' (novel) variation in the environment, for instance changes in habitat quality.




(a)




(b)

**Fig. 2.** (a) The logarithm of the time to population extinction against the value of $\log(K)/s$, where $K$ is the equilibrium population size (= $1/\alpha$ in Eqn 1) and $s$ is the standard deviation of log-abundances (base e). Results of numerical iterations of Eqn (1). Iterations were started at the equilibrium density and were continued until extinction or 1000 generations. The value of $s$ was estimated using data for generations 1 to 20 only, to reflect a realistic field situation. Results were obtained for the following combinations of model parameters: $r$ = 1, 1·5, 2; $\sigma$ = 0·2, 0·4, 0·6, 0·8, 1·0; $\alpha$ = 0·05, 0·025, 0·013, 0·006, 0·003; and $\theta$ = 0·3, 0·6, 0·9, 1·2, 1·5. (b) Shows the results in greater detail for $\log(K)/s < 5$; •, $\theta < 1$; ●, $\theta \geq 1$. ○, empirical data point for *Plebejus argus* (for details see the text).

Figure 3 shows the distribution of occupied patch sizes in the pooled data for five butterfly metapopulations (*T. acteon, P. argus, H. comma* and *M. athalia* from England and *M. cinxia* from Finland; for details see C. D. Thomas *et al.*, 1992). Of the 273 occupied patches none was >100 ha, and less than 5% were >10 ha (Fig. 3). Assuming a density of the order of 0·1 adult butterflies per m$^2$ (see Table 4 below), which is probably too high for many species, these figures suggest that local populations >10 000 individuals are

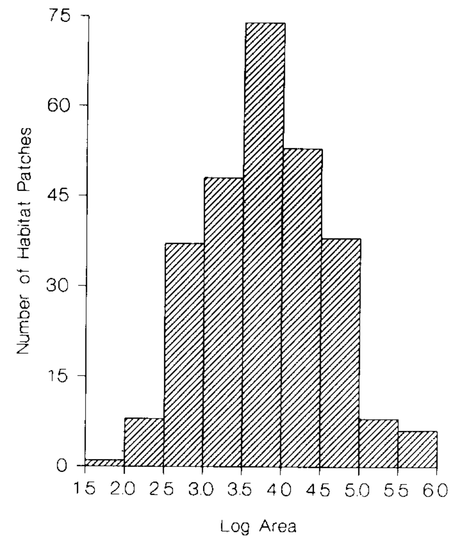



**Fig. 3.** Size distribution of occupied habitat patches in five butterfly metapopulations (*T. acteon*, *P. argus*, *H. comma* and *M. athalia* from England and *M. cinxia* from Finland; for details see C. D. Thomas *et al.*, 1992). Patch size is given in m$^2$ on a logarithmic scale (base 10). $n$ = 273.

uncommon for these butterflies, and that the vast majority of local populations are much smaller.

We conclude from the results in Figs 2 and 3 that many and perhaps most local populations of butterflies are within the size range at which they are clearly susceptible to 'normal' extinction. Even if some populations are now very large, unusual environmental events may occasionally drive them to extinction. For all these species it is necessary to expand the perspective from the level of local populations to that of metapopulations.

## A SPATIALLY EXPLICIT SIMULATION MODEL

The metapopulation model which we develop in this section is spatially explicit, i.e. a model which assumes certain sizes and spatial locations for habitat patches. The purpose of the model is to make quantitative predictions about the dynamics of specific metapopulations, including the transient dynamics following establishment (introduction), the number and kind of patches occupied at equilibrium, and persistence time. The model can be used to evaluate whether a particular set of habitat patches would support a metapopulation of a species with known biology; it can also be used to examine the consequences of various changes in the environment, for instance, a reduction in the average size of habitat patches or the elimination of some patches.

The model assumes discrete generations, as is the case in most of the rarer temperate butterflies. Local dynamics in habitat patch $i$ are modelled by the logistic equation

$$N_{i,t+1} = N_{i,t} \exp \{r[1 - (\alpha_i N_{i,t})^\theta]\} \quad (2)$$

which is the same as Eqn (1) but with no stochastic terms. $1/\alpha_i$ gives the equilibrium number of adult butterflies in patch $i$ in the absence of migration. In numerical iterations of the model, population size in each generation is truncated to the nearest integer value. The value of $1/\alpha_i$ can be estimated as the median density (adult butterflies per m$^2$) in existing local populations × the size of patch $i$, preferably excluding populations that have obviously been recently perturbed. The value of $r$ is estimated as the maximum yearly increase in population size, $\ln[N_{t+1}/N_t]$, when density is low, but not very low (the latter to avoid large sampling errors). The value of $\theta$ can be estimated with non-linear regression from data on yearly changes in population size, when there is no long-term temporal trend, using Eqn (2) with the estimated values of $\alpha_i$ and $r$. We omit stochastic terms in Eqn (2) for clarity and simplicity (we have constructed a version of the model with stochasticity in Eqn (2); it produced no qualitatively new results). Within-patch variability in population size is generated, in this model, by the processes of extinction, migration and colonization.

The model assumes that emigration is density- and patch size-independent, with fraction $c$ of individuals leaving their natal population in each generation. This assumption is unlikely to hold if there are huge differences in the sizes of habitat patches, because the rate at which individuals encounter the patch boundary would be much greater for small than large patches. We assume that immigration is independent of patch size and population size. This assumption too is a simplification, but it seems nonetheless a reasonable first approximation in the absence of detailed information on migration. *These assumptions highlight the need for good empirical data on emigration and immigration rates in butterflies.*

The number of individuals moving from patch $i$ to patch $j$ in generation $t$, $D_{ij,t}$, is estimated by

$$D_{ij,t} = cN_{i,t} \exp(-\mu d_{ij}) \frac{\exp(-\tau d_{ij})}{\sum \exp(-\tau d_{ij})} \quad (3)$$

where $d_{ij}$ is the distance between patches $i$ and $j$, and $e^{-\mu d_{ij}}$ is the fraction of individuals surviving migration over distance $d_{ij}$. Constant $\tau$ gives the shape of the distribution of migration distances; note that, ignoring mortality during migration, the distribution of immigrants per receiving patch against migration distance fits the negative exponential function. The values of $\tau$ and $\mu$ are difficult to estimate, because the observed distribution of migrants from patch $i$ to other patches is affected by both parameters. These parameters may need to be estimated with the model as described below.

The probability of successful colonization of an empty patch is given by parameter $\beta$, conditional on $\Sigma D_{j,t} > 0.5$. If females mate before migration, as is the case in many butterflies, $\beta$ could be set at 0·5 to reflect 50 : 50 sex ratio (if males are counted). If colonization is successful, the new population is started with $\Sigma D_{j,t}$ individuals, truncated to the nearest integer value.

The probability of local extinction of a population in patch $i$ in generation $t$ is given by

$$P_{i,t} = 1 - (1 - s_e)N_{i,t}^2/(s_d^2 + N_{i,t}^2) \quad (4)$$

**Table 2. The parameters of the simulation model**

| Process | Parameter | Description |
|---|---|---|
| Local growth | $r$ | Intrinsic rate of population increase |
| | $\theta$ | Strength of density dependence near equilibrium |
| | $\alpha$ | Inverse of carrying capacity (=population size at deterministic equilibrium) |
| Migration | $c$ | Fraction of individuals emigrating |
| | $\tau$ | Parameter setting the migration distances |
| | $\mu$ | Parameter setting the mortality rate of migrants |
| | $\beta$ | Probability of successful colonization |
| Extinction | $s_d$ | Demographic stochasticity (=population size at which extinction probability is about 0·5 per generation) |
| | $s_e$ | Environmental stochasticity (=extinction probability in very large populations) |

where the two parameters $s_e$ and $s_d$ reflect the levels of 'environmental' and 'demographic' stochasticities, affecting all populations and primarily small populations, respectively. $s_d$ gives the population size for which the per-year extinction probability is around 0·5, whereas $s_e$ gives the asymptotic per-year extinction probability in large populations. The value of $s_d$ can be estimated by scoring extinctions in many small, independent populations, and by fitting Eqn (4) to the data ($s_e$ may be set to 0 for this purpose). Parameter $s_e$ embodies the complex and long-term process of extinction in a large population. In practice, $s_e$ can probably be measured only using the simulation model (see below).

In summary, the model has nine parameters, of which three are associated with local dynamics, four are used to describe migration and colonization, and two the extinction process (Table 2). Ideally, one would like to have independent estimates of all nine parameters, make predictions about metapopulation dynamics for a particular set of habitat patches, and test the predictions with field observations. But as we have indicated above, at least $s_e$ may be practically impossible to estimate independently. When some parameter values are unknown, we suggest using the information on patterns of habitat patch occupancy and local density in some metapopulation at equilibrium to estimate, with the simulation model, the values of the unknown parameters. In this case, the model should naturally be validated by applying it with the same parameter values to independent data from some other metapopulation of the same species. This procedure will be demonstrated with examples in the next section.

## THREE CASE STUDIES

We illustrate the use of the simulation model with three examples from Finland, England and Wales. Independent estimates are not available for all model parameters, but all three examples do include the necessary information on the sizes and locations of all habitat patches, and the three studies include information, in various combinations, on local densities and transient metapopulation dynamics during a period of metapopulation expansion. We hope that these examples, apart from demonstrating the use and value of a spatially explicit model, would also give a clearer idea of what kind of information a field worker should attempt to gather.

### The three species

The Glanville fritillary, *Melitaea cinxia*, is widely but patchily distributed in Europe and Asia (Higgins & Riley, 1980; Emmet, 1990). It has become extinct from the British mainland, but has survived on the Isle of Wight, where it inhabits unstable cliffs (Simcox & Thomas, 1980; Pope, 1988; Emmet, 1990). The past distribution in Finland included the south-western archipelago between the mainland and the Åland islands, with one known population on the mainland (Marttila *et al.*, 1990). At present the species is almost completely restricted to the main Åland island, where it is most frequent on low rocky outcrops with naturally sparse vegetation (permanent habitat) and various dry meadows, including pastures (successional habitat).

The silver-spotted skipper, *Hesperia comma*, has a holarctic distribution (Higgins & Riley, 1980). In England, it is restricted to southerly-facing calcareous grasslands, where it lays its eggs on grasses (*Festuca ovina*) located adjacent to bare ground (J. A. Thomas *et al.*, 1986). Widespread at the turn of the century, *H. comma* declined with the ploughing of calcareous grasslands, and with reduced levels of grazing of remaining grasslands. It then became restricted to refuge populations when the disease myxomatosis killed rabbits in the mid-1950s, and the grasslands became overgrown. Recovery of rabbit populations has recreated habitat conditions needed by this species, but it has failed to recolonize many habitat patches that were historically populated. Given the specific habitat requirements of *H. comma* in Britain, we are able to identify habitat patches which are suitable but unoccupied, and so to map the distribution of vacant habitat patches (J. A. Thomas *et al.*, 1986; C. D. Thomas & Jones, 1993).

The silver-studded blue, *Plebejus argus*, is widely distributed in Europe and Asia but, as with *M. cinxia*, its distribution is patchy. *P. argus* has declined dramatically in Britain and it is now largely restricted to the south (C. D. Thomas, 1985*a,b*; Ravenscroft, 1990; C. D. Thomas & Harrison, 1992), with only a few northern populations surviving. On limestones in North Wales, *P. argus* inhabits southerly-facing grasslands which are subject to intermediate levels of grazing (C. D. Thomas, 1985*a,b*). If grazing is too heavy or too light, *P. argus* is eliminated. In this habitat, the main host plants are herbaceous Cistaceae (*Helianthemum chamaecistus* and *H. canum*) and Leguminosae (especially *Lotus corniculatus*). The eggs are laid along the margins between

**Table 3. Effects of habitat patch isolation (measured by index *I*, Eqn 5) and area ($\log_{10}$-transformed) on patch occupancy (presence/absence) and density (individuals per ha; $\log_{10}$-transformed)**

Results from multiple logistic regression (occupancy) and multiple linear regression (density). Units of area and distance are ha and km, respectively. Density estimates were collected in one year for each species.

| | | | *M. cinxia* | *H. comma*[a] | *P. argus*[b] |
|---|---|---|---|---|---|
| Occupancy | Isolation | Coeff. | −0·0058 | −0·0035 | Negative[c] |
| | | t | 1·37 | 3·00 | |
| | | P | 0·179 | 0·004 | <0·001 |
| | Area | Coeff. | 1·25 | 2·51 | 2·85 |
| | | t | 2·24 | 3·67 | 2·51 |
| | | P | 0·030 | <0·001 | 0·018 |
| | | $n_1$[d] | 50 | 67 | 31 |
| Density | Isolation | Coeff. | −0·0262 | −0·0100 | −0·001 |
| | | t | 2·75 | 2·95 | 1·37 |
| | | P | 0·009 | 0·005 | 0·207 |
| | Area | Coeff. | −0·690 | −0·163 | 0·177 |
| | | t | 11·25 | 1·77 | 0·72 |
| | | P | <0·001 | 0·084 | 0·490 |
| | | $n_2$[d] | 42 | 50 | 11 |
| | | $R^2$ | 0·76 | 0·23 | 0·24 |

[a] Region 1a.
[b] Region 1a plus 1b.
[c] There was a complete separation of occupied versus empty patches in terms of isolation because Region 1b was empty (Fig. 8(c)). Isolation was omitted from the model.
[d] $n_1$, number of suitable habitat patches; $n_2$, number of occupied habitat patches.

bare ground and vegetation, and the immature stages have a mutualistic association with *Lasius* ants (C. D. Thomas, 1985*a*; Ravenscroft, 1990; Jordano & Thomas, 1992; Jordano *et al.*, 1992). As with the two previous butterfly species, it is relatively easy (with experience) to recognize suitable but unpopulated habitat patches (C. D. Thomas & Harrison, 1992).

### Effects of isolation and patch area

It is straightforward, in principle, to measure habitat patch areas and local population densities, but patch isolation is a more complex variable. There is no single measure of isolation that would be obviously superior to many others. Following Hanski *et al.* (in press), we have measured the isolation of patch *i* using the following index

$$I_i = -\sum \exp(-d_{ij})N_j/100 \qquad (5)$$

where $N_j$ is population size in patch *j* (divided by 100 for numerical convenience) and $d_{ij}$ is the distance between patches *i* and *j* (in km). The minus sign is used to give a positive correlation between the level of isolation and the value of $I_i$. The sum is taken over all patches except patch *i*. This measure of isolation is affected by all occupied patches, with their effects being weighted by distance from patch *i*, and by their population sizes. Other constants than 1 could be used in the exponent in Eqn (5), to give more or less weight to long distances, but the results would be qualitatively similar. For simplicity, we have used the value of 1 throughout this paper.

**Table 4. Statistics on local densities in the three species of butterflies**

| Local density (individuals $m^{-2}$) | *M. cinxi* | *H. commas*[a] | *P. argus*[b] |
|---|---|---|---|
| Mean | 0·401 | 0·183 | 0·198 |
| Median | 0·094 | 0·120 | 0·100 |
| Minimum | 0·008 | 0·012 | 0·004 |
| Maximum | 4·416 | 0·870 | 0·860 |
| n[c] | 42 | 50 | 27 |

[a] Region 1a.
[b] Regions pooled.
[c] n is the number of local populations.

Table 3 summarizes the results for the three species in the study areas described below. The effect of isolation on patch occupancy is negative in all three species, and the effect is significant in two species (exception *M. cinxia*, perhaps because there were only a few isolated patches). In all three cases no clearly outlying, empty habitat patches were included; such patches would naturally generate a negative relationship. The effect of patch area on occupancy is positive and significant in all three species (Table 3). These results are consistent with the predictions of simple patch models of metapopulation dynamics (Hanski, 1991, 1994), in which increasing isolation is assumed to decrease colonization rate, and hence make occupancy less likely, and in which increasing area decreases extinction rate, making occupancy more likely.

In two species, *M. cinxia* and *H. comma*, local density decreased significantly with increasing isolation and with increasing patch area (only *M. cinxia* significant). *P. argus* showed no significant effect of isolation or area on local density. The effect of isolation in *M.*

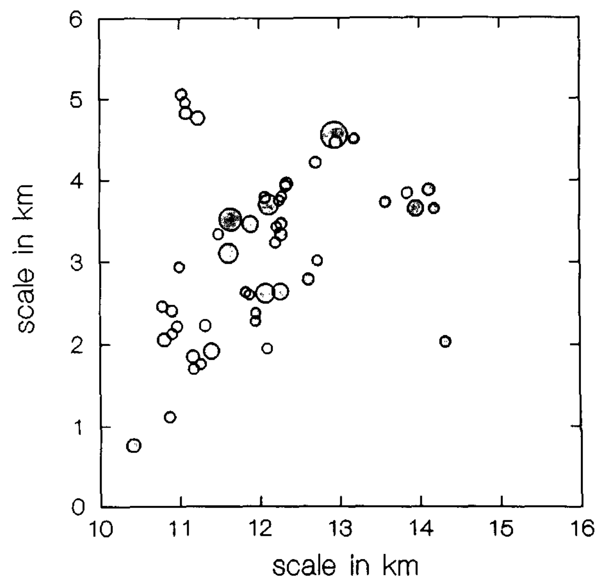



**Fig. 4.** Locations and relative sizes of the 50 habitat patches in the *Melitaea cinxia* metapopulation. ●, patches occupied in 1991. Note that the areas of the circles indicate only the relative sizes of the habitat patches (many patches are so small that they would be hardly visible if shown in the scale of the map, the patch sizes ranged from 12 $m^2$ to 4·6 ha; see Hanski *et al.*, in press).

**Table 5. The final parameter values for the three species of butterflies (for the parameters see Table 2 and the text)**

| Parameter | *M. cinxia* | *H. comma* | *P. argus* |
|---|---|---|---|
| $r$ | (2·0) | **1·5** | **2·0** |
| $\theta$ | 0·1 | 0·2 | 1·0 |
| $\alpha^{1}$ | **0·1** | **0·1** | **0·1** |
| $c$ | **0·3** | 0·1 | 0·1 |
| $\tau$ | **1·0** | 1·0 | 2·0 |
| $\mu$ | 0·25 | 0·4 | 0·5 |
| $\beta$ | (0·5) | (0·5) | (0·5) |
| $s_d$ | 20 | 32 | 32 |
| $s_e$ | 0·006 | 0·008 | 0·02 |

Values in brackets are based on the literature (in *Mellicta athalia*, a close relative of *M. cinxia*, $r$ = 2·1; Warren, 1991; the $\beta$ values assume that half of the immigrants are mated females). Values given in bold face were estimated using mark–recapture and other methods (unpublished data). The other values were estimated indirectly by fitting the model to the data as described in the text.

*cinxia* and *H. comma*, and lack of a significant effect in *P. argus*, is most parsimoniously explained by the effect of migration on local dynamics (and lack of this effect in *P. argus*), which may also generate the negative relationship between area and density in the first two species.

Table 4 gives statistics on local density. The median density is around 0·1 butterflies per m$^2$ in all three species. The very high maximum value for *M. cinxia* is probably an artefact of an extremely small minimum patch size (12 m$^2$).

### Predicted and observed metapopulation dynamics in *Melitaea cinxia*

Hanski *et al.* (in press) studied a *M. cinxia* metapopulation in an area of 15 km$^2$ on the main Åland island, southern Finland, with 50 habitat patches suitable for *M. cinxia* (Fig. 4). The habitat patches consisted mostly of low, rocky outcrops, typically surrounded by cultivated fields and forest. The group of 50 patches was well separated from other groups of habitat patches. *M. cinxia* has one generation per year and its principal larval host plant is *Plantago lanceolata* (Hanski *et al.*, in press).

An intensive mark–recapture study provided estimates of emigration rate and realised migration distances (Hanski *et al.*, in press). A value for the intrinsic growth rate was obtained from the literature (on the related species *Mellicta athalia*; Warren, 1991), and the probability of successful colonization was assumed to be 0·5, because most migrating females are mated. This leaves four parameters unknown. We estimated the values of these parameters by running the simulation model with a large number of parameter combinations and selecting the combination which generated metapopulation patterns most closely matching the observed ones (for details see Appendix 1). Table 5 gives the 'best-matching' parameter values, and Table 6 shows a comparison between the predicted and observed metapopulation patterns.

The match between the predictions and observations is surprisingly good, both in the magnitude and in the statistical significance of the effects of patch isolation and area on occupancy and local density (Table 6). The parameter values which generated the 'best-matching' prediction are consistent with our intuitive expectations, except perhaps the value of $\theta$ (Table 5), which sets the degree of density dependence near equilibrium. The value $\theta$ = 0·1 indicates weak density dependence. It was apparent from the simulation results that a low value of $\theta$ was needed to obtain the observed steep relationship between local density and patch area (Table 6).

Using the parameter values given in Table 5, we calculated for each of the 50 patches the predicted frequency of being occupied at equilibrium, after 100 generations (Fig. 5). Eighteen patches were occupied in each of the 10 replicate simulations, and of them only one was empty, in reality, in 1991. The remaining seven patches which were empty in 1991 did not show any clear relationship to the model prediction (Fig. 5), suggesting that there is substantial randomness in patch occupancy in this metapopulation.

**Table 6. Comparison between the observed and predicted effects of isolation and patch area on patch occupancy and local density**
The values are multiple regression coefficients ($t$-values in brackets). The predicted values were obtained using the parameter values given in Table 5.

| | | *M. cinxia* | | *H. comma* | | *P. argus*[b] | |
|---|---|---|---|---|---|---|---|
| | | Observed | Predicted[a] | Observed | Predicted | Observed | Predicted |
| Occupancy | Isolation | −0·005 | −0·011 | −0·005 | −0·009 | −0·001 | −0·008 |
| | | (1·23) | (1·84) | (3·93) | (3·44) | (7·94) | (9·63) |
| | Area | 0·062 | 0·030 | 0·145 | 0·089 | 0·102 | 0·135 |
| | | (2·44) | (1·18) | (5·21) | (4·11) | (1·49) | (2·86) |
| Density | Isolation | −0·026 | −0·030 | −0·010 | −0·007 | −0·001 | −0·001 |
| | | (2·75) | (6·04) | (2·95) | (2·48) | (1·37) | (0·58) |
| | Area | −0·690 | −0·540 | −0·163 | −0·186 | 0·177 | 0·276 |
| | | (11·25) | (26·58) | (1·77) | (7·21) | (0·72) | (2·76) |
| Fraction of patches occupied | | 0·84 | 0·86 | 0·75 | 0·85 | 0·35 | 0·36 |

[a] The predicted values are means for generations 91 to 100, when the simulated metapopulation had reached an equilibrium.
[b] Group 1a.

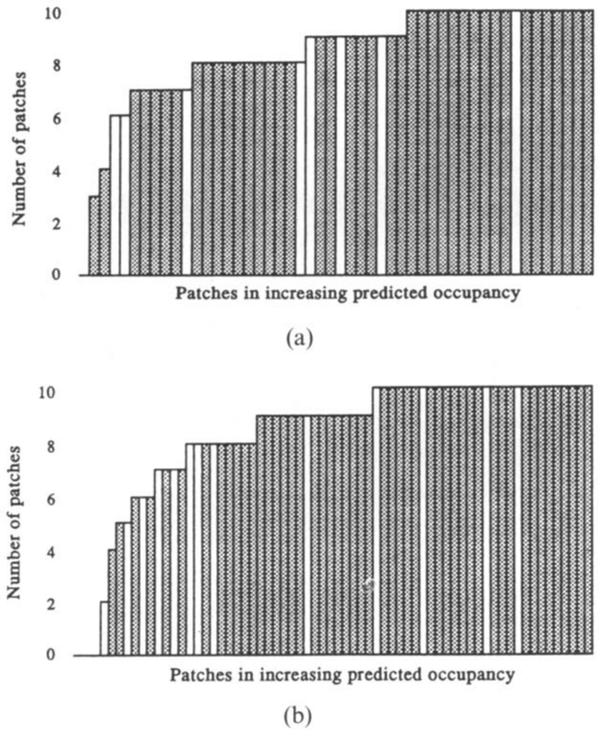



**Fig. 5.** The number of simulations out of 10 in which a particular habitat patch was occupied at equilibrium (after 100 generations) in (a) the *M. cinxia* and (b) the *H. comma* metapopulations. The patches have been arranged, along the horizontal axis, in the order of increasing predicted frequency of occupancy. Patches shown by white bars were empty in reality.

We have not attempted any sensitivity analyses of the model results because of the large number of parameters for which there is no independent estimate. Although the results in Table 5 must be interpreted cautiously, simulation results clearly showed that substantially different parameter values would predict entirely different metapopulation patterns.

**Predicted and observed metapopulation dynamics in *Hesperia comma***

C. D. Thomas and Jones (1993) report data for two networks of suitable habitat patches for *H. comma* in South England. The two systems of patches are so isolated from each other that the respective metapopulations function independently (Fig. 6). In region 1a in Surrey and W Kent, with 67 patches, *H. comma* is widespread and assumed to be close to equilibrium (Fig. 6(a); absent in region 1b with 27 patches). In region 2 in E Sussex (Fig. 6(b)), with 75 patches, *H. comma* declined during the early part of the century, and it was found in only one large and two small patches in a survey conducted in 1982. Subsequently habitat quality has improved due to increasing grazing by rabbits, and the species has started to spread in region 2a (C. D. Thomas & Jones, 1993).

Unfortunately, there are no independent estimates for any of the model parameters apart from the equilibrium local density and $r$. We therefore ran the model with a large number of parameter combinations, and used the comparison between the observed and predicted metapopulation patterns to select the 'best-matching' set of parameter values (for details see Appendix 1). Clearly, this procedure is far from ideal because of the large number of unknown parameters, and further work should be directed towards estimating the parameter values with independent observations or experiments. However, as the parameter estimates were obtained by fitting the model to data from region 1 only, we may test model predictions with independent data from region 2.

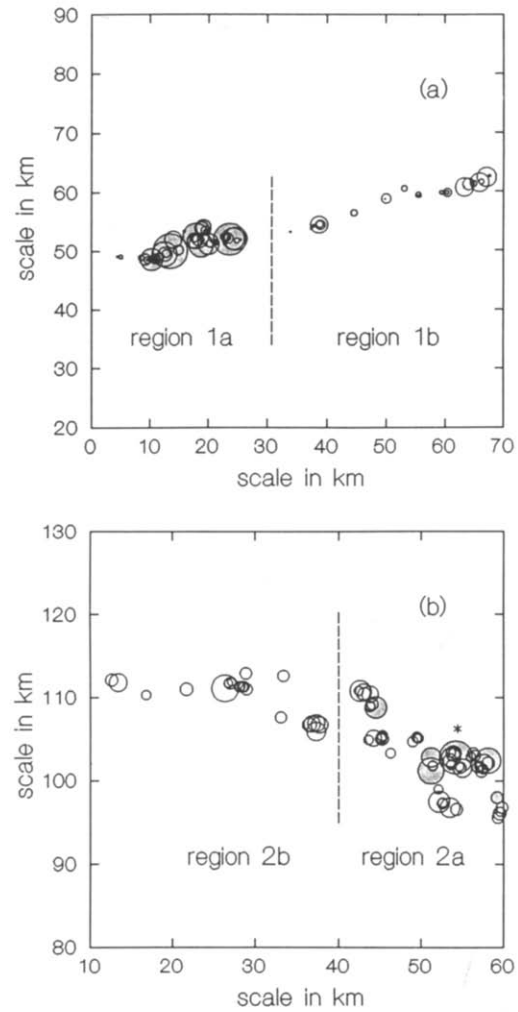



**Fig. 6.** Locations and relative sizes of (a) the 94 (region 1) and (b) 75 (region 2) habitat patches in two *Hesperia comma* metapopulations. The broken lines split the regions into two subregions, discussed in the text. ●, occupied patches in 1991 (note that all patches in regions 1b and 2b were empty). In (b) the asterisk (*) indicates the large habitat patch which was occupied, with two small, nearby patches, in 1982 and from which the rest of the shaded patches were colonized by 1991. The areas of the circles indicate only the relative sizes of the habitat patches (many patches are so small that they would be hardly visible if shown in the scale of the map; for more details see C. D. Thomas & Jones, 1993).

The set of 'best-matching' parameter values (Table 5) are feasible for *H. comma*, and they again predict very well the effects of habitat patch area and isolation on patch occupancy and local density (Table 6). As with *M. cinxia*, we ran 10 replicate simulations to calculate the predicted frequency of occupancy for all the patches (Fig. 5). Most of the patches (13 out of 17) that were empty in reality had a predicted frequency of occupancy less than one, though as with *M. cinxia*, some patches which were expected to be occupied were, in reality, empty, indicating substantial randomness in patch occupancy (Fig. 5).

Region 1b with 27 patches has remained unoccupied, apparently because of their isolation from the rest of the patches (Fig. 6 (a)). In the model, region 1b was not colonized naturally from the rest of the patches, but if the butterfly was 'experimentally' (by simulation) introduced to region 1b, it often became established. If the species was initially placed in about 10% of patches (with probability 0·1 in each patch), it went rapidly extinct in two out of 10 trials, and occupied three to 12 patches at equilibrium, after 100 generations, in the remaining eight trials. If the species was introduced initially to about 90% of patches (with probability 0·9 in each patch), no introduction failed, and the equilibrium number of patches occupied was 10 to 17 patches in 10 replicate trials. Figure 6 shows that the 27 patches in region 1b include a cluster of relatively large patches and several smaller and more isolated ones, most of which were empty at equilibrium. The interesting point is that such patch configurations lead to alternative, 'metastable' patterns of patch occupancy in the entire metapopulation.

We next used the parameter values given in Table 5 to run the model for region 2, assuming that in year 0 *H. comma* was restricted to the three patches in which it was observed in 1982, and calculating the number of patches occupied after nine years (generations), when another survey was conducted (1991) and 21 patches were found to be occupied (Fig. 6(b)). In all but one of 100 replicate simulations the establishment was successful, and a mean of 8·6 patches were occupied after nine years (median 9, minimum 1, maximum 11). This is significantly less than the observed number, 21. After 50 generations 14·4 patches were occupied, still significantly less than the observed number after nine years, though there was a good agreement between prediction and observation in which patches were occupied (Fig. 7).

There are two likely reasons for the discrepancy between the predicted and observed numbers of occupied patches after nine years. First, patch occupancy in region 1a, which was used to estimate parameter values, was perhaps not at equilibrium in 1991, but the metapopulation may have been expanding, as the one in region 2 did (there was indeed some spread from 1982 to 1991, and a drought year in 1990 may have temporarily reduced patch occupancy below the equilibrium level). This would give parameter values underestimating the species' colonization potential. Secondly, it may be difficult to predict correctly, with the present model, long-distance migration, which is critical during rapid metapopulation expansion. In region 2, two patches had been colonized by 1991 though they had zero probability of colonization in the model even after 50 years (Fig. 7). These patches

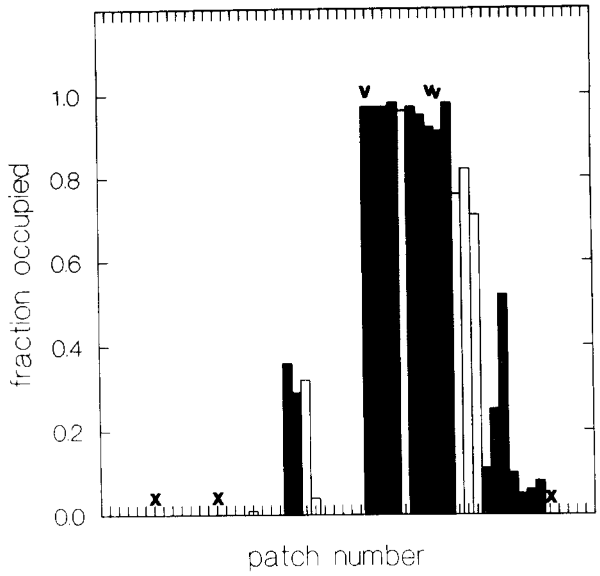



**Fig. 7.** The numbers of simulations out of 100 in which the 54 habitat patches in region 2a in Fig. 6 were occupied after 50 years. Each simulation was run using the parameter values given in Table 5 and assuming that in year 0 the three local populations shown by 'v' in the figure were at their equilibrium size. These patches were occupied, in reality, in 1982. ■, patches occupied, in reality, in 1991; x, patches which were occupied in reality but in none of the 100 simulations. The 54 patches have been arranged along the horizontal axis in an order roughly corresponding to their locations from west to east in Fig. 6(b).

were isolated by up to 8·65 km from the nearest occupied patch, indicating some potential for long-range migration and colonization.

A subset of 21 more isolated patches in region 2 (region 2b in Fig. 6(b)) has in reality so far remained uncolonized and it was never colonized in the simulations in 100 generations. 'Experimental' introductions by simulation to region 2b in the model gave similar results to those obtained for region 1b (above). If region 2b was colonized from a few patches (with probability 0·1 in each patch), the colonization failed in five out of 10 trials, and in the remaining trials four to eight patches were occupied after 100 years. When colonization occurred simultaneously in many patches (with probability 0·9 in each patch), colonization was always successful, and 10 to 18 patches were occupied after 100 years.

### Predicted and observed metapopulation dynamics in *Plebejus argus*

The study area in North Wales has 70 habitat patches in three groups suitable for *P. argus* (Fig. 8). *P. argus* has occurred naturally in one group of patches (region 1, Fig. 8(c)), but by 1983 it had gone extinct in the subgroup shown as region 1b in Fig. 8(c). *P. argus* was introduced into the remaining two networks of patches in 1942 (Fig. 8(b)) and in 1983 (Fig. 8(a)).

Only two parameter values can be estimated independently: $\alpha$, which sets the equilibrium density, and $r$, the intrinsic rate of population increase. Additionally, we assume a value of 0·5 for $\beta$, the colonization proba-

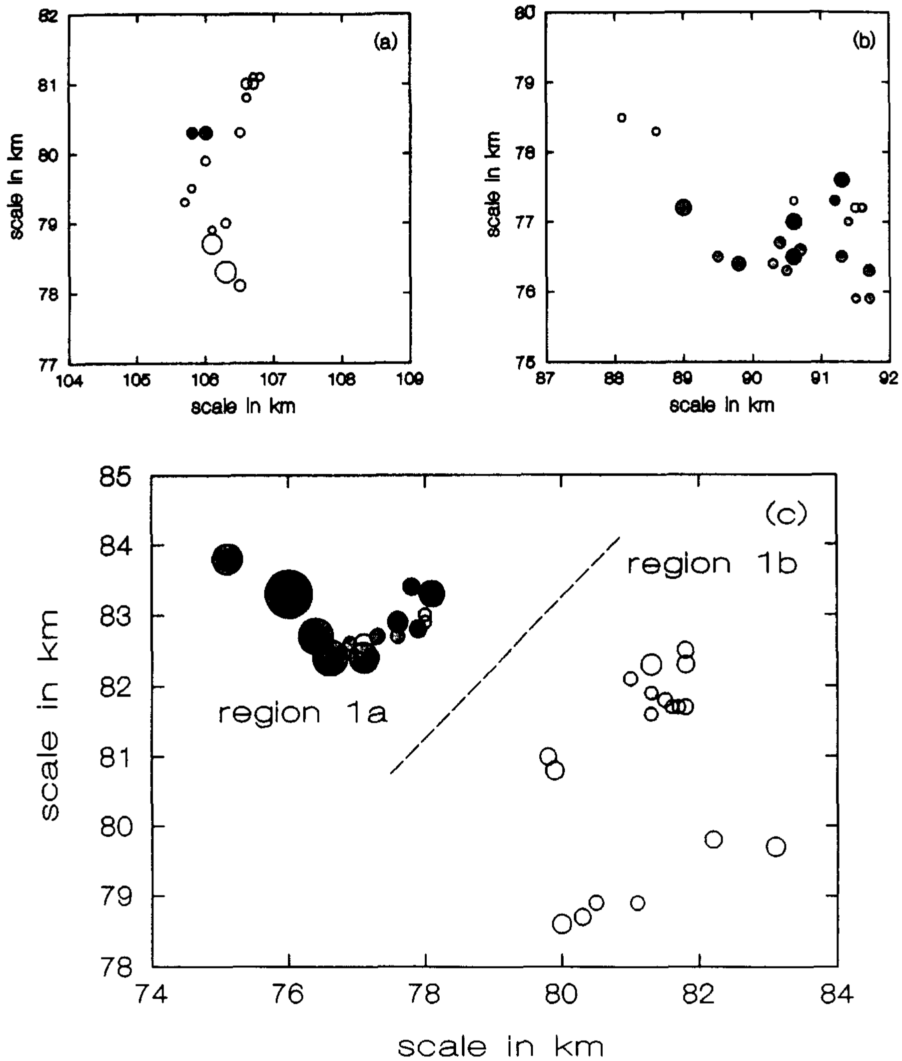



**Fig. 8.** Locations and relative sizes of 70 habitat patches in three *Plebejus argus* metapopulations within a study area in North Wales (region 3 in (a), region 2 in (b) and region 1 in (c)). The broken line splits region 1 into two subregions, discussed in the text. ●, patches occupied in the year of the survey (note that all patches in region 1b were empty). The areas of the circles indicate only the relative sizes of the habitat patches (many patches are so small that they would be hardly visible if shown in the scale of the map; for more details see C. D. Thomas & Harrison, 1992).

bility, as in the other species (Table 5). The remaining parameter values were estimated by fitting the model to the observed data (Fig. 8(c)). This approach did not yield a satisfactory result, however, because several combinations of parameters gave roughly equal fit. We therefore abandoned the plan of testing model predictions with data from Figs 8(a) and 8(b), and instead used these data to select parameter values predicting observations for all three groups of patches. Although these results cannot be tested with independent data, it remains interesting to find out whether a single set of parameter values can explain the observations for all three regions.

The final parameter values are given in Table 5. With the patches in Fig. 8(c), we started a simulation with most patches occupied. All populations in region 1b, where *P. argus* has gone extinct in reality, went extinct in the model by generation 260. From this point onwards, only region 1a was occupied, and the predicted effects of patch area and isolation on occupancy and local density are in general agreement with observations (Table 6). With the patches in Fig. 8(b), we started 100 simulations by assuming that the patch into which *P. argus* was introduced in 1942 was occupied. In reality, 17 patches were occupied after 48 years, in 1990 (Fig. 8(b)). In the 100 simulations, the introduction failed in 34 cases, in 47 cases less than 11 patches were occupied, and in 19 cases more than 10 patches were occupied after 48 years (maximum 18). The observed metapopulation spread is consistent with these results.

With the patches in Fig. 8(a), 100 simulations were run for seven years by assuming that the patch into which *P. argus* was introduced in 1983 was occupied. In reality, only two patches were occupied seven years later, in 1990 (Fig. 8(a)). In the 100 simulations, the introduction failed in 10 cases, whereas the mean number of patches occupied in the remaining 90 cases was 2·7 (median 2, minimum 1 and maximum 7). Thus in this case too the observation is consistent with model predictions.

In summary, these results are encouraging by demonstrating that the same set of parameter values correctly predicts the ultimate extinction of *P. argus* in the current patches in region 1b (Fig. 8(c)); the observed metapopulation patterns in region 1a (Table 6); and the transient dynamics in the two other networks of habitat patches (Fig. 8(a),(b)). The next obvious step is to test model predictions with independent data.

## DISCUSSION

### Butterfly metapopulation dynamics

Our results demonstrate frequent turnover of local populations in three species of temperate butterflies, *Melitaea cinxia, Hesperia comma* and *Plebejus argus*. In all three species, the effects of habitat patch area and isolation on patch occupancy and local density are consistent with predictions based on metapopulation theory. The three examples represent metapopulations which are not dominated by a large 'mainland' population, and differ in this respect from the *Euphydryas editha* metapopulation studied in California by Harrison *et al.* (1988).

Based on the results of the simulation model, the most notable differences between the three species appear to be in migration distances and in the level and type of density dependence. Compared with *M. cinxia* and *H. comma, P. argus* is predicted to have shorter migration distances, and stronger density dependence around the equilibrium (long-term average) population size. The predicted difference in migration distances is consistent with mark–recapture data (C. D. Thomas, 1985*a*; J. A. Thomas *et al.*, 1986; Hanski *et al.*, in press) and patterns of patch occupancy and colonization (C. D. Thomas & Harrison, 1992; C. D. Thomas, 1993; C. D. Thomas & Jones, 1993; Hanski *et al.*, in press). A more surprising result was the higher value of $\theta$ in *P. argus* than in the two other species, indicating weak density dependence near equilibrium in *M. cinxia* and *H. comma*, but nearly linear density dependence in *P. argus*. Weak density dependence is consistent with Dempster's (1983) findings for 14 species of butterflies. The value of $\theta$ in the model is largely determined by the effect of patch area on local density (Table 3). This relationship would also be affected by density-dependent migration, or area-dependent emigration. We assumed, in the model, density- and patch size-independent migration in the absence of contrary results, but this question needs to be re-addressed when more detailed data on migration become available.

The ecological significance of the observed differences between the species can be assessed by running the model for each metapopulation with the parameter values estimated for the other species (Table 7). As the parameter values themselves (Table 5) suggest, *M. cinxia* had the highest while *P. argus* had the lowest frequency of patch occupancy in the different sets of habitat patches (*cinxia*, small $s_d$ and small $\mu$, *argus* large $s_d$, $\mu$ and $\tau$). All species survived 100 generations in all systems of patches, when the simulation was started with most patches occupied, as in Table 7. The next examples illustrate, however, that a species may fail to invade the same network of patches in which it persists for at least 100 generations if present in most patches initially.

**Table 7. Predicted metapopulation size as measured by the fraction of patches occupied in the three species of butterflies using the parameter values for different species (from Table 5)**
The values given here are the mean fractions of patches occupied in generation 100 in 10 replicate simulations, starting with all patches occupied.

| Parameters of | Patch structure of | | |
|---|---|---|---|
| | *M. cinxia* | *H. comma*[a] | *P. argus*[b] |
| *M. cinxia* | 0·85 | 0·94 | 0·94 |
| *M. comma* | 0·40 | 0·84 | 0·66 |
| *P. argus* | 0·36 | 0·76 | 0·62 |

[a] Region 1a.
[b] Region 1a.

Figure 9 shows the predicted change in the fraction of occupied patches after 100 generations in hypothetical situations with all patch areas or isolations being changed by a constant factor. The results are shown for two starting configurations of occupied patches, namely the observed patterns with most patches occupied (as in Table 7), and only a small subset (about 10%) of the patches occupied. The fraction of patches occupied after 100 generations depends on the initial set of occupied patches (Fig. 9), because the decline from high occupancy is necessarily quite slow, due to a relatively low extinction rate in the largest patches. The results indicate that the *M. cinxia* metapopulation would not remain viable if patch areas were reduced only slightly, to less than 80% of their present sizes (Fig. 9). The *Hesperia comma* metapopulation would tolerate the greatest reduction in patch areas, as it presently has the largest patches. Increasing patch isolation would be most detrimental to the *P. argus* metapopulation (Fig. 9), apparently because of its low migration rate.

### Butterfly extinctions

Local extinctions have been frequent among British butterflies (J. A. Thomas, 1984, 1991; Warren, 1993*a*), and long-term persistence of many species most probably depends on genuine colonization–extinction dynamics in networks of many habitat patches. We suggest that this is not an unusual situation in scores of temperate butterflies in the often highly fragmented landscapes (C. D. Thomas *et al.*, 1992; C. D. Thomas, 1993).

Although it is clear that the main cause of extinctions in butterflies has been deteriorating habitat quality, and that extinctions are therefore impossible to predict by standard population dynamic models, it is also true that small populations tend to be more extinction-prone than large ones, as predicted by theoretical models (Fig. 2). C. D. Thomas and Harrison (1992) surveyed habitat patches occupied by *Plebejus argus* seven years (generations) apart. In small patches, less than 0·2 ha in size, they found 80% turnover (extinc-

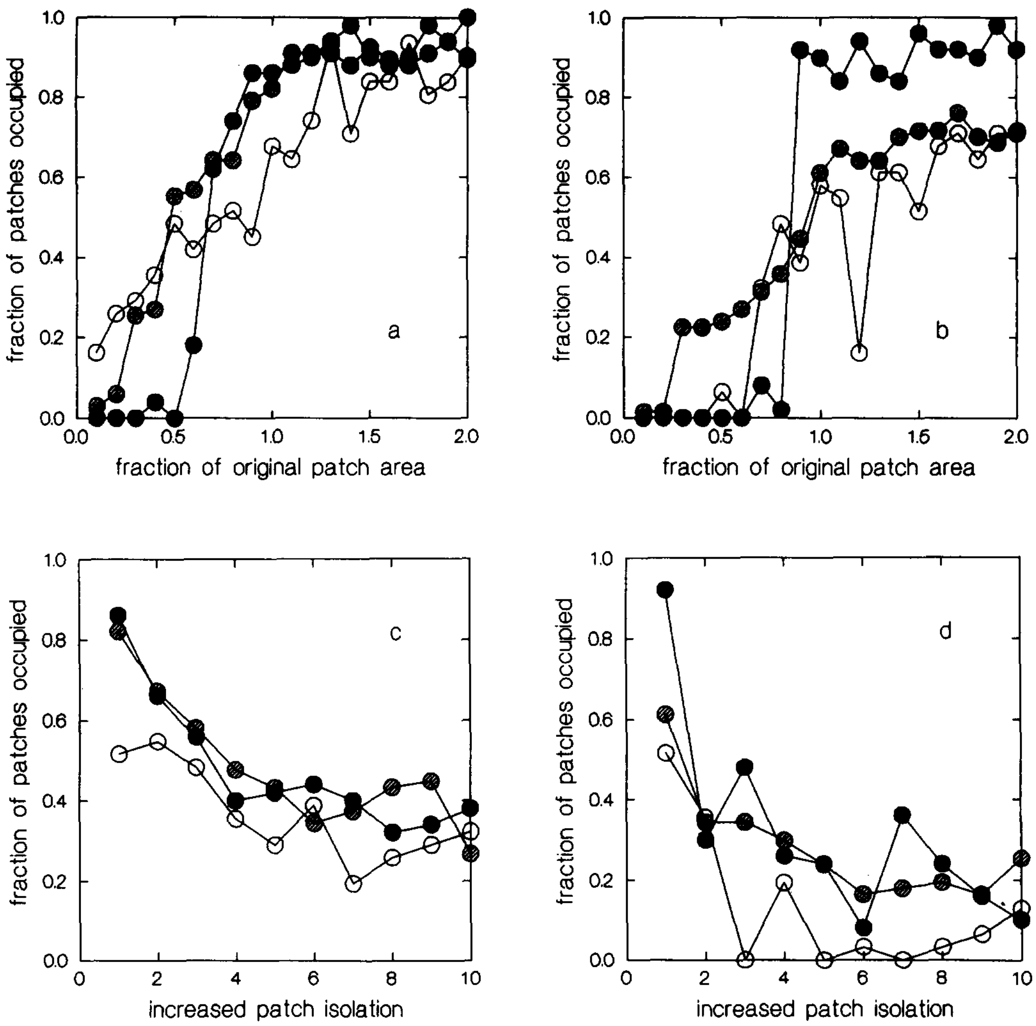



**Fig. 9.** Predicted changes in the fraction of occupied patches in generation 100 in metapopulations of *M. cinxia* (●), *H. comma* (◍) and *P. argus* (○) when the areas of all the patches were changed by a constant factor (a), (b) or their isolations were increased by a constant factor (c), (d). Figure 9(a) and (c) shows the results of simulations which were started with most patches occupied, whereas Fig. (b) and (d) shows the results of simulations started with a smaller fraction of occupied patches.

tions and colonizations), in medium size patches, from 0·2 to 0·9 ha, turnover was 35%, whereas in larger patches turnover was just 10%. For *Hesperia comma*, the probability of extinction over a nine-year period was similarly greatest for small and relatively isolated patches (C. D. Thomas *et al.*, 1992; C. D. Thomas & Jones, 1993). The Californian checkerspot *Euphydryas editha* became extinct on several relatively small habitat patches, following a '50-year drought', but it survived on a 'mainland' patch (Harrison *et al.*, 1988).

It is worth stressing that, for many temperate butterflies, the size distribution of habitat patches (Fig. 3) is such that there is no mainland site at which populations would be safe from extinction. For these species local populations typically lie in the size range in which extinction is a real threat, and long-term persistence must be seen in a metapopulation context.

**Butterfly introductions**

A number of butterfly introductions have been spectacularly successful. *Plebejus argus* was released in the Dulas valley in Wales in 1942, and it is still thriving after 50 years, having populated an entire network of habitat patches (Fig. 8(b); C. D. Thomas, 1985*b*, C. D. Thomas & Harrison, 1992). Another British rarity, *Strymonidia pruni*, was introduced to Surrey in 1952, and has spread up to 4 km from the original release point (J. A. Thomas & Lewington, 1991; C. D. Thomas *et al.*, 1992). The spread of *Hesperia comma* in East Sussex represents a natural reinvasion of a network of previously unsuitable but presently suitable patches from a refuge patch (Fig. 6(b)).

The very successful introductions of *P. argus* and *S. pruni*, and the natural reinvasion of *H. comma*, demonstrate that introductions and re-establishments can be highly successful in the long term, and also that suitable habitat often remains unoccupied because of isolation. Successful introductions have generally occurred to networks of vacant habitat patches, suggesting that introductions should be planned and executed in the metapopulation context rather than, as has been the common practice so far, at the scale of single habitat patches and single local populations (C. D. Thomas, 1992). Most introductions to single habitat patches in the UK have eventually failed, even if a local population was established for a number of years (reviewed by

Oates & Warren, 1990). Although we would not expect all entomologists and conservationists to construct specific metapopulation models before making releases, much could be gained by comparing the characteristics of networks of vacant habitat patches with those that are occupied by a conspecific metapopulation. In some cases, it may be possible to create 'stepping-stone' habitat patches to help species bridge the gap between distant networks of patches.

### Butterfly conservation

An obvious but important advantage of spatially explicit metapopulation models of the type described here is that they can be used to make quantitative predictions about the likely course of events in a given species in a given set of habitat patches. For instance, is species *A* likely to persist in the set of habitat patches *X*? Would a (re)introduction of species *A* into region *Y* result in a viable metapopulation? Would extra management that increased the size, number, or quality of patches improve the situation? Or conversely, if 50% of the total area will be lost, would it be less harmful for metapopulation persistence to lose the largest, or the smallest, patches, accounting for 50% of total area, than 50% of the area of each patch?

Because the results of numerical simulations of the kind outlined above and exemplified by Fig. 9 are necessarily specific to particular species in particular patches, further studies are needed before it will be possible to make more general conclusions and conservation recommendations. Nevertheless, it is clear that the larger the number of large patches within colonization distance of one another the better. Less obvious is that events occurring outside protected areas—say decreased patch number or increased distances between patches—may greatly affect the ability of a species to persist *within* a protected area. *Events that take place in the wider (non-reserved) countryside may be as relevant to long-term persistence as those that take place on small reserves.* An integrated approach is needed in which conservation management and habitat protection is directed at networks of habitat patches.

We conclude by suggesting that spatially explicit metapopulation models of the type described here provide a useful, unified framework in which to analyse and compare results for particular metapopulations. These models also hold the promise of becoming a practical and cost-effective management tool for butterfly conservation and for conservation of metapopulations in general. From the management point of view, we emphasize the value of using a metapopulation model to compare two or more alternative management options. For most species, it would be expensive (and often impossible) to obtain accurate parameter estimates and to test the model exhaustively, but if the aim is to choose between alternative options we really need only qualitatively, not quantitatively, correct predictions. We suggest that the kind of model and data described in this paper would generally be sufficient for this purpose.

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## APPENDIX 1: NUMERICAL SIMULATIONS

The predicted and observed metapopulation patterns were matched by minimizing the values of $y$, $x_1$ and $x_2$, defined as

$$y = \text{abs}\,(n_o - n_p)$$
$$x_1 = \text{abs}\,(b_{1o} - b_{1p})/b_{1o} + \text{abs}\,(b_{2o} - b_{2p})/b_{2o}$$
$$x_2 = \text{abs}\,(b_{3o} - b_{3p})/b_{3o} + \text{abs}\,(b_{4o} - b_{4p})/b_{4o}$$

where $n_o$ and $n_p$ are the observed and predicted number of occupied patches, and $b_{io}$ and $b_{ip}$ are the observed and predicted regression slopes of patch occupancy versus isolation ($i$ = 1) or patch area ($i$ = 2), and patch density versus isolation ($i$ = 3) or patch area ($i$ = 4). The regression slopes for occupancy and density were calculated with multiple regressions, in the same way in both the simulated and real data. The predicted number of occupied patches and the predicted regression slopes are the means for generations 91 to 100, when the model metapopulation had settled to equilibrium. The parameter combination which gave the smallest sum of $x_1$ and $x_2$, with small $y$ (usually less than 5), was selected.

For *M. cinxia*, the following combinations of parameters were tried in independent simulations (without replication): $\theta$ = 0·1, 0·2, 0·4, 0·8, 1·6; $\mu$ = 0·25, 0·5, 1, 2, 4; $s_d$ = 5, 10, 20, 40, 80; and $s_e$ = 0·003, 0·006, 0·012, 0·024, 0·048. The remaining parameters were kept constant as shown in Table 5. For *H. comma* the following combinations of parameters were used: $\theta$ = 0·05, 0·1, 0·2, 0·4; $c$ = 0·05, 0·1, 0·2, 0·4; $\tau$ = 0·5, 1, 2, 4; $\mu$ = 0·1, 0·2, 0·4, 0·8; $s_d$ = 4, 8, 16, 32; and $s_e$ = 0·002, 0·008, 0·032, 0·128. The remaining parameters were kept constant as shown in Table 5. For *P. argus* see the text.